

DEERWOOD, MB, Canada

WMO: 718400

Lat: 49.3997N

Lon: 98.3192W

Elev: 341

StdP: 97.29

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.2	-26.1	-32.0	0.2	-28.1	-29.8	0.2	-25.9	14.2	-13.1	13.1	-12.9	5.2	280	0.698

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	30.2	21.2	28.5	20.4	26.9	19.3	23.4	27.9	22.0	26.7	20.8	25.1	4.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	17.3	26.2	20.5	15.8	24.9	19.1	14.5	23.6	71.7	28.1	66.3	26.7	61.7	25.3	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.9	9.8	DB	-31.7	34.1	2.1	1.7	-33.2	35.3	-34.4	36.4	-35.6	37.3	-37.1	38.6	(4)
(5)			WB	-31.8	25.5	2.2	1.7	-33.3	26.8	-34.6	27.8	-35.8	28.7	-37.4	30.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-12.8	-12.9	-5.1	4.0	11.0	17.0	19.8	19.0	14.1	5.9	-2.8	-11.8
	DBStd	13.42	8.09	7.21	7.28	6.15	5.03	3.57	3.15	3.35	4.67	5.18	6.96	7.42
	HDD10.0	3292	708	642	470	193	49	1	0	0	16	149	387	677
	HDD18.3	5447	967	875	727	429	233	65	21	33	142	385	635	936
	CDD10.0	1051	0	0	1	15	79	211	304	278	137	22	2	0
	CDD18.3	165	0	0	0	0	6	25	67	53	14	1	0	0
	CDH23.3	1546	0	0	0	3	79	237	559	509	151	7	0	0
	CDH26.7	393	0	0	0	0	18	48	134	155	36	2	0	0
Wind	WSAvg	4.8	5.6	5.3	5.0	5.0	5.2	4.3	3.9	4.0	4.3	5.0	5.1	5.1
Precipitation	PrecAvg	536	16	16	23	28	79	97	79	65	47	42	28	26
	PrecMax	683	35	54	63	57	166	177	199	141	103	156	87	56
	PrecMin	412	1	2	0	6	30	31	16	22	6	7	2	3
	PrecStd	79	8	12	16	15	37	39	39	32	21	31	21	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	5.3	16.4	23.3	29.1	30.2	31.9	33.0	30.3	23.8	16.7	5.2
		MCWB	3.1	2.6	11.0	13.1	16.0	22.1	24.0	22.4	18.6	15.2	11.2	2.5
	2%	DB	2.8	2.8	10.8	19.7	25.6	27.9	29.7	30.3	27.2	19.8	12.0	2.7
		MCWB	1.1	1.0	7.2	10.7	14.0	19.1	22.6	20.9	18.0	12.4	6.9	0.7
	5%	DB	0.9	0.5	6.5	17.1	22.5	26.0	28.1	28.3	24.9	16.7	9.1	0.6
		MCWB	-0.4	-0.6	4.0	9.3	13.2	18.3	21.7	20.3	17.1	10.9	5.6	-0.7
	10%	DB	-1.1	-2.0	3.8	14.2	19.8	24.2	26.6	26.2	22.1	14.0	6.4	-1.5
		MCWB	-2.0	-3.0	2.0	8.0	12.2	17.5	20.4	18.9	15.4	9.8	3.8	-2.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	2.8	11.8	14.7	18.2	23.9	25.8	24.9	21.7	16.1	10.8	2.9
		MCDB	5.1	5.0	14.5	19.7	25.2	28.0	29.7	29.6	27.6	22.1	16.6	5.3
	2%	WB	1.3	1.2	7.7	11.2	15.9	21.6	23.9	22.6	19.5	13.8	7.7	0.8
		MCDB	2.7	2.6	11.3	17.8	20.8	25.5	27.9	28.0	24.0	18.2	11.1	2.5
	5%	WB	-0.4	-0.4	4.5	9.7	14.6	19.7	22.6	21.2	17.9	11.7	5.8	-0.7
		MCDB	0.8	0.8	6.9	15.8	20.0	23.9	26.9	26.2	22.7	15.9	8.8	0.5
	10%	WB	-2.4	-2.8	2.4	8.2	13.3	18.3	21.3	19.9	16.5	10.0	3.9	-2.5
		MCDB	-1.4	-1.8	3.9	13.6	18.7	22.6	25.5	24.6	21.0	14.0	6.3	-1.7
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.6	8.5	11.1	11.6	10.5	11.0	11.9	11.5	9.4	7.8	7.4
		MCDBR	9.5	9.5	10.7	15.9	15.7	13.5	12.5	14.7	15.2	13.6	11.6	9.5
		MCWBR	8.7	7.8	8.0	9.0	7.7	7.2	7.3	7.2	7.7	7.8	7.9	8.0
	5% WB	MCDBR	9.7	9.1	10.4	14.7	12.4	11.6	11.9	12.9	13.4	12.6	11.0	9.0
		MCWBR	8.7	7.8	7.8	8.9	7.4	7.3	7.7	7.4	8.0	7.7	7.7	7.9
Clear-Sky Solar Irradiance	taub	0.270	0.285	0.297	0.357	0.373	0.378	0.385	0.382	0.347	0.299	0.268	0.250	
	taud	2.259	2.239	2.381	2.321	2.338	2.355	2.345	2.344	2.431	2.541	2.482	2.382	
	Ebn at Noon	780	859	920	888	886	881	868	854	852	849	801	773	
	Edh at Noon	72	94	98	116	120	119	118	113	93	70	58	55	
All-Sky Solar Radiation	RadAvg	1.48	2.67	3.92	4.99	5.54	5.96	6.33	5.19	3.79	2.31	1.31	1.05	
	RadStd	0.12	0.21	0.32	0.36	0.43	0.41	0.27	0.34	0.28	0.30	0.18	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	-0.68	N/A	N/A	N/A	N/A

Nomenclature: See separate page